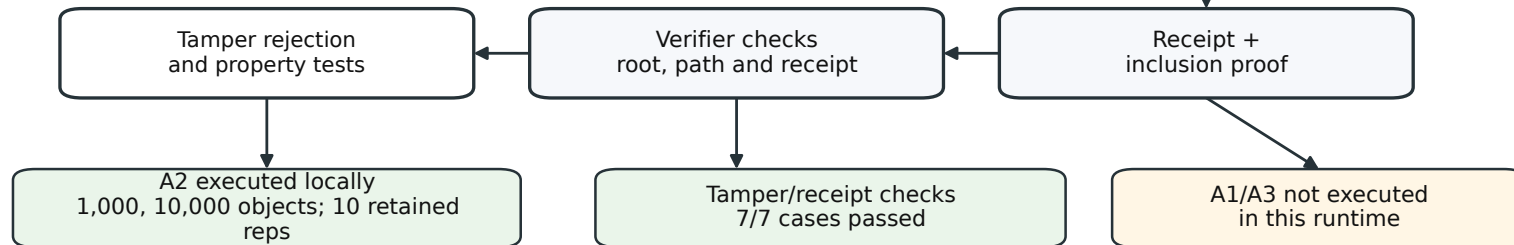


A2 Merkle local reference execution and verification flow

append path



verification path



Local Python reference implementation only; not production performance or external witnessing.